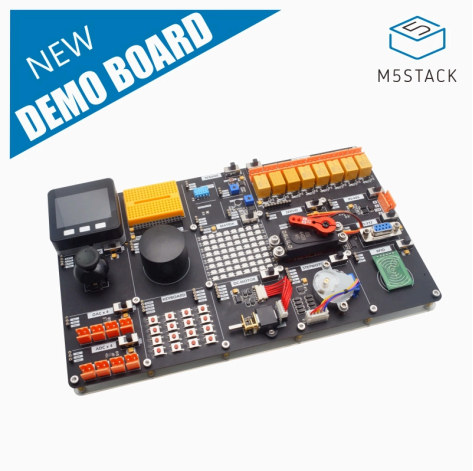
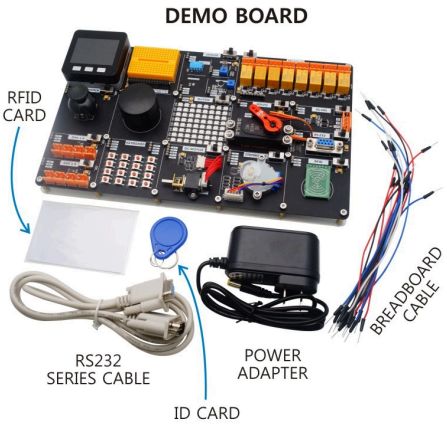
Demo Board

SKU:K024







Description

Demo Board is a learning and development board. It uses the M5Core as the control core and is fully compatible with the module stacking and hardware expansion system. It is equipped with multiple sets of environment detection-related sensors and provides various input methods such as joystick, rotary encoder, matrix buttons, and wireless radio frequency identification. It includes three motor driving methods (DC, stepper, and servo), an RGB LED panel, integrated multiple sets of relay control, and ADC and DAC conversion circuits. It supports RS485 and RS232 bus communication and provides an independent power switch for each module. Combined with the IoT-enabled M5Core as the control core, the onboard modules cover multiple aspects of "sound, light, electricity, and force," making the Demo Board a powerful tool for learning hardware and programming.

Features

- Compatible with Module stacking and Unit expansion systems
- Proto board and M5-BUS bus expansion
- Independent power switch for each module
- Environmental sensor series (temperature, humidity, barometric pressure, light, microphone)
- Joystick input
- 8-channel relay output
- 4-channel DAC, 4-channel ADC
- 4x4 button matrix
- 8x8 matrix RGB LED
- Rotary encoder
- Single-channel servo
- DC motor (with feedback)
- Four-phase five-wire stepper motor
- Wireless radio frequency identification card reader
- RS-485, RS232 communication functions

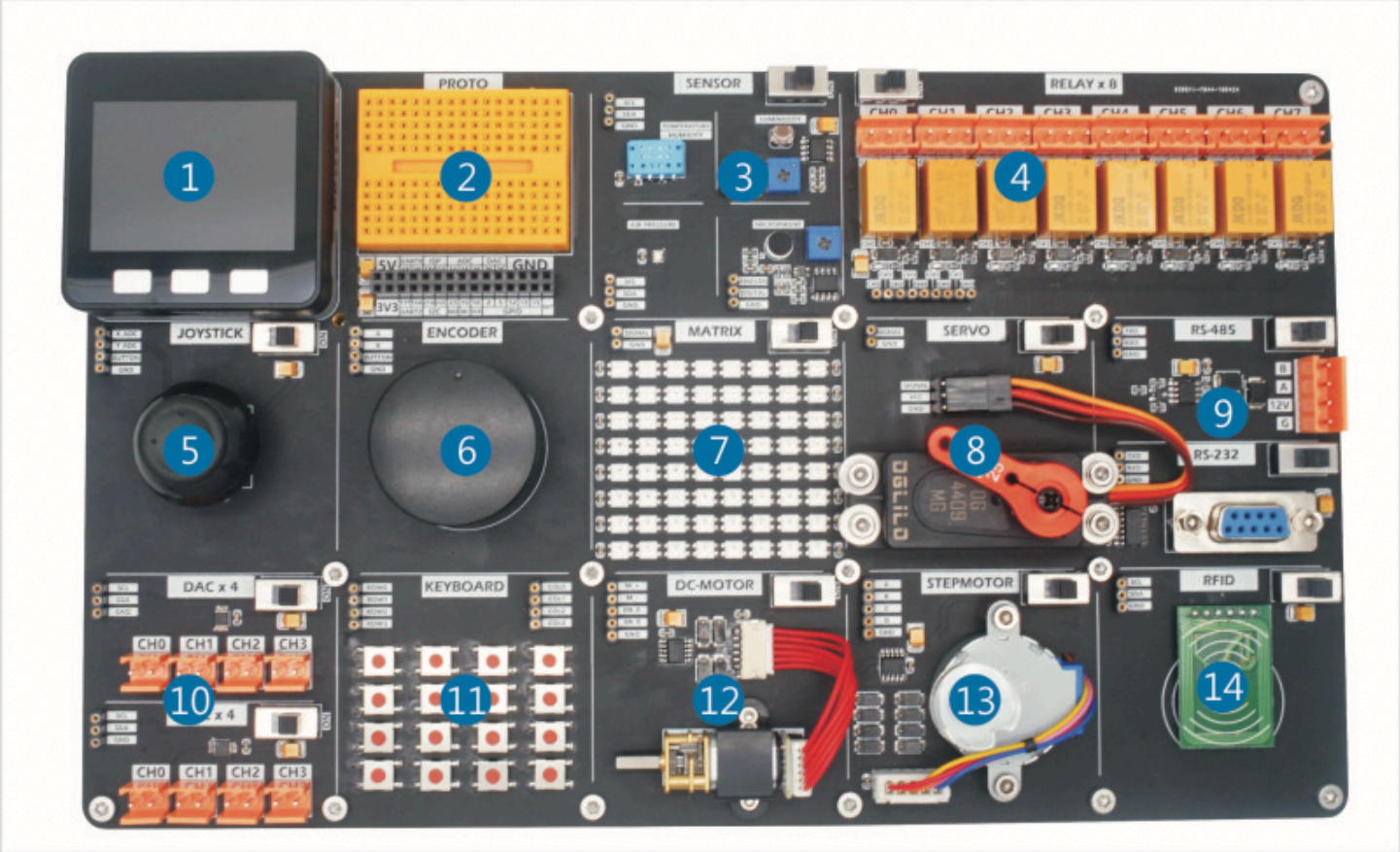
Includes

- 1 x Demo Board
- 1 x DC 12V power adapter (5.5 x 2.1mm)
- 1 x RS232 cable
- 1 x RFID Card
- 1 x ID Card

- 16 x Breadboard wires

Learn

Module Layout



1: CORE 2: PROTO 3: SENSOR 4: RELAYx8

5: JOYSTICK 6: ENCODER 7: MATRIX 8: SERVO

9: RS-485/RS-232 10: DACx4/ADCx4 11: KEYBOARD

12: DC-MOTOR 13: STEPMOTOR 14: RFID

Module Parameters

Module Name	Operating Voltage	Related Parameters
ADC	5V	4-channel ADC interface / built-in ADS1115
DAC	5V	4-channel DAC interface / built-in DAC6574
Joystick	3.3V	X/Y-axis potentiometer input, Z-axis button input
DHT12	3.3V	I2C address 0x5C
BMP280	3.3V	I2C address 0x76
Light	3.3V	Supports analog/digital acquisition / adjustable threshold
Microphone	3.3V	Supports analog/digital acquisition / adjustable threshold
Relay	5V	8-channel control / 3A-220V-AC / 3A-30V-DC
RGB LED	5V	8x8 matrix LED
Servo	5V	10KG torque
DC-Motor	5V	With feedback, integrated LV8548MC
Stepmotor	5V	Four-phase five-wire, integrated LV8548MC
RFID	3.3V	Read/write distance: < 8 cm / built-in MFRC522
RS485	5V	Built-in SP485EEN-L/TR
RS232	5V	Built-in MAX232ESE
Encode		Rotary encoder / with button input
Proto		170 board holes
Keyboard		4x4 button matrix

Schematics

- Demo Board Schematics PDF

Datasheets

- ADS1115
- DAC6574
- LV8548MC
- TPS54360
- MRC522
- MAX232ESE
- MAX4466
- SP485EEN
- BMP280

Softwares

Quick Start

- Demo Board Quick Start Guide PDF

Arduino

- Demo Board Test Example

Easyloader

Easyloader	Download Link	Notes
Demo Board Test Easyloader	download	/

Note: The DAC and ADC functions cannot be used simultaneously because the ADS1115 and DAC6574 chips share the same broadcast address (0x48), causing a functional conflict.